

## 11.3 Testable hypotheses

Scientific questions are usually phrased in terms of coefficients rather than subspaces: “the coefficients of poverty and single parenthood are equal”, “the four regions have the same intercept”, “the elasticity is one”. Such statements have the form

$$H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d},$$

with a known  $p \times m$  matrix  $\mathbf{\Lambda}$  and a known  $\mathbf{d} \in \mathbb{R}^m$ . This is the **general linear hypothesis**. This section decides when such a statement is a hypothesis about the distribution of  $\mathbf{Y}$  at all, gives its  $F$  statistic in terms of the unrestricted fit alone, shows that the statistic agrees with the reduced-model statistic of [Section 11.1](#), and describes least squares under the hypothesis. Nothing here assumes that  $\mathbf{X}$  has full column rank.

### 11.3.1 Which constraints are hypotheses

In the full model the distribution of  $\mathbf{Y}$  depends on  $\boldsymbol{\beta}$  only through  $\mathbf{X}\boldsymbol{\beta}$ . A statement about  $\boldsymbol{\beta}$  that cannot be read off from  $\mathbf{X}\boldsymbol{\beta}$  is therefore not a statement about the data-generating distribution, and no data can test it.

#### Definition 11.3.1 — Testable hypothesis

Let  $\mathbf{\Lambda}$  be a  $p \times m$  matrix and  $\mathbf{d} \in \mathbb{R}^m$ . The hypothesis  $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  is **testable** if

- (i)  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$  is estimable, that is,  $\mathcal{C}(\mathbf{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$  (Definition 8.2.1 and Theorem 8.3.1), so that  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$  for some  $n \times m$  matrix  $\mathbf{T}$ ; and
- (ii) the equations  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$  are consistent, that is,  $\mathbf{d} \in \mathcal{C}(\mathbf{\Lambda}^\top)$  (Theorem 1.4.1).

Its number of degrees of freedom is  $q = \text{rank}(\mathbf{\Lambda})$ . We assume  $q \geq 1$ .

Condition (ii) is automatic when  $\mathbf{\Lambda}$  has full column rank, for then  $\mathcal{C}(\mathbf{\Lambda}^\top) = \mathbb{R}^m$ . It matters when the columns of  $\mathbf{\Lambda}$  are redundant, as when all six pairwise differences of four group effects are set to zero. Redundant constraints are harmless if they agree with each other. If they do not, as in  $\alpha_1 - \alpha_2 = 1$ ,  $\alpha_2 - \alpha_3 = 1$ ,  $\alpha_1 - \alpha_3 = 0$ , the “hypothesis” is false for every  $\boldsymbol{\beta}$ , and no test is possible.

Condition (i) is the substantive one. Without it the constraint cannot be settled by any amount of data.

#### Proposition 11.3.1 — Nonestimable constraints cannot be tested

Let  $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$  and  $c \in \mathbb{R}$ .

- (a) For every  $\boldsymbol{\beta} \in \mathbb{R}^p$  there is  $\boldsymbol{\beta}'$  with  $\mathbf{X}\boldsymbol{\beta}' = \mathbf{X}\boldsymbol{\beta}$  and  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}' = c$ .
- (b) Let  $\phi$  be any test of  $H : \boldsymbol{\lambda}^\top \boldsymbol{\beta} = c$  in the model Eq. 7.3.1, with rejection probability  $\pi(\boldsymbol{\beta}, \sigma^2) = \mathbb{E}_{\boldsymbol{\beta}, \sigma^2} \phi(\mathbf{Y})$ . For every point  $(\boldsymbol{\beta}, \sigma^2)$  of the alternative there is a point  $(\boldsymbol{\beta}', \sigma^2)$  of the hypothesis with  $\pi(\boldsymbol{\beta}', \sigma^2) = \pi(\boldsymbol{\beta}, \sigma^2)$ . So no test has power

exceeding its size anywhere.

*Proof.*

- (a) By Theorem 8.3.1(c),  $\lambda \notin \mathcal{C}(\mathbf{X}^\top)$  means that  $\lambda$  is not orthogonal to  $\mathcal{N}(\mathbf{X})$ : there is  $\mathbf{v}$  with  $\mathbf{X}\mathbf{v} = \mathbf{0}$  and  $\lambda^\top \mathbf{v} \neq 0$ . Put  $\beta' = \beta + t\mathbf{v}$  with  $t = (c - \lambda^\top \beta)/\lambda^\top \mathbf{v}$ . Then  $\mathbf{X}\beta' = \mathbf{X}\beta$  and  $\lambda^\top \beta' = c$ . (b) The law of  $\mathbf{Y}$ , and hence  $\pi$ , depends on  $(\beta, \sigma^2)$  only through  $(\mathbf{X}\beta, \sigma^2)$ . Apply (a). ■

For a matrix  $\Lambda$ , Proposition 9.2.6 gives the general statement: the constraint  $\Lambda^\top \mathbf{b} = \mathbf{0}$  restricts the mean vector only through its estimable part  $\mathcal{C}(\Lambda) \cap \mathcal{C}(\mathbf{X}^\top)$ , and the extra sum of squares has  $\dim(\mathcal{C}(\Lambda) \cap \mathcal{C}(\mathbf{X}^\top))$  degrees of freedom. In the one-way layout  $E(Y_{ij}) = \mu + \alpha_i$ , the constraint  $\alpha_1 = 0$  is a *side condition* (Theorem 8.4.2). It selects one of the equivalent parameterizations and says nothing about the data. Programs that report a “test” of  $\alpha_1 = 0$  are testing something else, usually a contrast such as  $\alpha_1 - \alpha_g$  that the program’s coding makes estimable.

### 11.3.2 The F statistic of a testable hypothesis

The sum of squares of a testable hypothesis was computed in Theorem 9.2.4 for  $\Lambda$  of full column rank. The next theorem removes that restriction, identifies the reduced model, and supplies the distribution.

#### Theorem 11.3.2 — The $F$ test of a testable hypothesis

Assume Eq. 7.3.1 with  $r < n$ , and let  $H : \Lambda^\top \beta = \mathbf{d}$  be testable with  $q = \text{rank}(\Lambda)$ . Let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ ,  $\hat{\beta}$  any least squares estimate,  $\mathbf{W} = \Lambda^\top \mathbf{G} \Lambda$  and  $\mathbf{W}^-$  any generalized inverse of  $\mathbf{W}$ . Put

$$\text{SS}_H = (\Lambda^\top \hat{\beta} - \mathbf{d})^\top \mathbf{W}^- (\Lambda^\top \hat{\beta} - \mathbf{d}), \quad F_H = \frac{\text{SS}_H/q}{\text{SSE}/(n-r)}. \quad (11.3.1)$$

- (a)  $\text{SS}_H$  does not depend on the choices of  $\hat{\beta}$ ,  $\mathbf{G}$  and  $\mathbf{W}^-$ .  
 (b) Let  $\mathbf{b}_0$  be any solution of  $\Lambda^\top \mathbf{b}_0 = \mathbf{d}$  and  $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \Lambda^\top \mathbf{b} = \mathbf{0}\}$ . Then  $\mathcal{S}_0$  is a subspace of  $\mathcal{C}(\mathbf{X})$  of dimension  $r - q$ , the set of mean vectors allowed by  $H$  is  $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ , and

$$\text{SS}_H = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 - \text{SSE} = \|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})(\mathbf{y} - \mathbf{X}\mathbf{b}_0)\|^2.$$

- (c)  $F_H \sim F(q, n - r, \gamma)$  with

$$\gamma = \frac{(\Lambda^\top \beta - \mathbf{d})^\top \mathbf{W}^- (\Lambda^\top \beta - \mathbf{d})}{\sigma^2}, \quad (11.3.2)$$

and  $\gamma = 0$  iff  $\Lambda^\top \beta = \mathbf{d}$ . In particular  $F_H$  is the  $F$  statistic of Theorem 11.1.1 for the data  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$  and the reduced model space  $\mathcal{S}_0$ .

*Proof. Reduction to full column rank.* Choose  $q$  linearly independent columns of  $\mathbf{\Lambda}$ , forming a  $p \times q$  matrix  $\mathbf{\Lambda}_1$ . Every column of  $\mathbf{\Lambda}$  is a combination of them, so  $\mathbf{\Lambda} = \mathbf{\Lambda}_1 \mathbf{K}$  for a  $q \times m$  matrix  $\mathbf{K}$ , which has rank  $q$  because  $\text{rank } \mathbf{\Lambda} = q$ . Hence  $\mathbf{K} \mathbf{K}^\top$  is nonsingular. Put  $\mathbf{d}_1 = \mathbf{\Lambda}_1^\top \mathbf{b}_0$ , so that  $\mathbf{d} = \mathbf{K}^\top \mathbf{d}_1$ . Since  $\mathbf{K}^\top$  has full column rank it is one-to-one, so  $\mathbf{d}_1$  is determined by  $\mathbf{d}$  and does not depend on the choice of  $\mathbf{b}_0$ . Moreover  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d} \iff \mathbf{K}^\top (\mathbf{\Lambda}_1^\top \mathbf{b} - \mathbf{d}_1) = \mathbf{0} \iff \mathbf{\Lambda}_1^\top \mathbf{b} = \mathbf{d}_1$ . The same argument with  $\mathbf{d}$  replaced by  $\mathbf{0}$  shows that  $\mathcal{S}_0$  is also  $\{\mathbf{X} \mathbf{b} : \mathbf{\Lambda}_1^\top \mathbf{b} = \mathbf{0}\}$ . The columns of  $\mathbf{\Lambda}_1$  are columns of  $\mathbf{\Lambda}$ , so  $\mathbf{\Lambda}_1^\top \boldsymbol{\beta}$  is estimable.

(a). By Theorem 9.2.4(a),  $\mathbf{u}_1 = \mathbf{\Lambda}_1^\top \hat{\boldsymbol{\beta}} - \mathbf{d}_1$  and the positive definite matrix  $\mathbf{W}_1 = \mathbf{\Lambda}_1^\top \mathbf{G} \mathbf{\Lambda}_1$  do not depend on  $\hat{\boldsymbol{\beta}}$  and  $\mathbf{G}$ . Now  $\mathbf{u} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d} = \mathbf{K}^\top \mathbf{u}_1$  and  $\mathbf{W} = \mathbf{K}^\top \mathbf{W}_1 \mathbf{K}$ . Let  $\mathbf{v} = \mathbf{K}^\top (\mathbf{K} \mathbf{K}^\top)^{-1} \mathbf{W}_1^{-1} \mathbf{u}_1$ , so that  $\mathbf{K} \mathbf{v} = \mathbf{W}_1^{-1} \mathbf{u}_1$ . Then  $\mathbf{W} \mathbf{v} = \mathbf{K}^\top \mathbf{W}_1 \mathbf{K} \mathbf{v} = \mathbf{K}^\top \mathbf{u}_1 = \mathbf{u}$ , and for every generalized inverse,

$$\mathbf{u}^\top \mathbf{W}^- \mathbf{u} = \mathbf{v}^\top \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{v} = \mathbf{v}^\top \mathbf{W} \mathbf{v} = (\mathbf{K} \mathbf{v})^\top \mathbf{W}_1 (\mathbf{K} \mathbf{v}) = \mathbf{u}_1^\top \mathbf{W}_1^{-1} \mathbf{u}_1.$$

The right side involves none of the choices.

(b).  $\mathcal{S}_0$  is the image of the subspace  $\mathcal{N}(\mathbf{\Lambda}^\top)$  under  $\mathbf{b} \mapsto \mathbf{X} \mathbf{b}$ , so it is a subspace of  $\mathcal{C}(\mathbf{X})$ . Its dimension is  $r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top)) = r - q$  by Proposition 9.2.6, because  $\mathcal{C}(\mathbf{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ . The solutions of  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$  are  $\mathbf{b}_0 + \mathcal{N}(\mathbf{\Lambda}^\top)$ , so the allowed means are  $\mathbf{X} \mathbf{b}_0 + \mathcal{S}_0$ . For the minimum, write  $\mathbf{y}_* = \mathbf{y} - \mathbf{X} \mathbf{b}_0$ . As  $\mathbf{b}$  ranges over the constraint set,  $\mathbf{X} \mathbf{b} - \mathbf{X} \mathbf{b}_0$  ranges over  $\mathcal{S}_0$ , so

$$\min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \min_{\mathbf{m} \in \mathcal{S}_0} \|\mathbf{y}_* - \mathbf{m}\|^2 = \|(\mathbf{I} - \mathbf{P}_{\mathcal{S}_0}) \mathbf{y}_*\|^2$$

by Theorem 6.3.1. Also  $\text{SSE} = \|(\mathbf{I} - \mathbf{M}) \mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M}) \mathbf{y}_*\|^2$ , since  $\mathbf{X} \mathbf{b}_0 \in \mathcal{C}(\mathbf{X})$ . Because  $\mathcal{S}_0 \subseteq \mathcal{C}(\mathbf{X})$ , Eq. 6.5.2 turns the difference into  $\|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0}) \mathbf{y}_*\|^2$ . By Theorem 9.2.4(c) applied to  $\mathbf{\Lambda}_1$  and  $\mathbf{d}_1$ , the same difference equals  $\mathbf{u}_1^\top \mathbf{W}_1^{-1} \mathbf{u}_1$ , which is  $\text{SS}_H$  by (a).

(c).  $\mathbf{Y} - \mathbf{X} \mathbf{b}_0 \sim N_n(\mathbf{X}(\boldsymbol{\beta} - \mathbf{b}_0), \sigma^2 \mathbf{I})$  is a normal linear model with mean in  $\mathcal{C}(\mathbf{X})$ , and by (b)  $F_H$  is its  $F$  statistic for the reduced model space  $\mathcal{S}_0$ , which has dimension  $r - q$ . Theorem 11.1.1 gives  $F_H \sim F(q, n - r, \gamma)$  with  $\sigma^2 \gamma = \|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})(\mathbf{X} \boldsymbol{\beta} - \mathbf{X} \mathbf{b}_0)\|^2$ . To evaluate this, note that the identity in (b) holds for every data vector. Apply it to the vector  $\mathbf{X} \boldsymbol{\beta}$ , for which  $\boldsymbol{\beta}$  itself is a least squares estimate:  $\sigma^2 \gamma = (\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top \mathbf{W}^- (\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})$ . Finally, by (a) this equals  $\mathbf{u}_1(\boldsymbol{\beta})^\top \mathbf{W}_1^{-1} \mathbf{u}_1(\boldsymbol{\beta})$  with  $\mathbf{u}_1(\boldsymbol{\beta}) = \mathbf{\Lambda}_1^\top \boldsymbol{\beta} - \mathbf{d}_1$ . It is zero iff  $\mathbf{\Lambda}_1^\top \boldsymbol{\beta} = \mathbf{d}_1$ , that is, iff  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ . ■

Formula Eq. 11.3.1 needs only the unrestricted fit. Its ingredients have a statistical meaning:  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  is the estimated departure from the hypothesis, and  $\sigma^2 \mathbf{W}$  is its covariance matrix (Theorem 7.5.1(a)). So  $\text{SS}_H / \sigma^2$  is the squared Mahalanobis length of the estimated departure, and the noncentrality Eq. 11.3.2 is the same length for the true departure. Under  $H$  the quadratic form  $\text{SS}_H / \sigma^2$  is  $\chi^2(q)$ , which is Corollary 7.5.3. The theorem adds the reduced model, the distribution under alternatives, and the fact that a redundant  $\mathbf{\Lambda}$  can be used as it stands, with any generalized inverse of  $\mathbf{W}$ .

The test space of the hypothesis is the orthogonal complement of  $\mathcal{S}_0$  within  $\mathcal{C}(\mathbf{X})$ . By Theorem 6.5.2 it is  $\mathcal{C}(\mathbf{M} \mathbf{T})$ , where  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ : the hypothesis constrains the mean vector to be orthogonal to the fitted values of the columns of  $\mathbf{T}$ .

### 11.3.3 Least squares under the hypothesis

The minimum in [Theorem 11.3.2\(b\)](#) is attained, and its minimizers have a closed form. The following extends [Theorem 9.2.5](#), which assumed full rank, to every model matrix.

#### Proposition 11.3.3 — Least squares under a testable hypothesis

Let  $H : \Lambda^\top \beta = d$  be testable with  $\Lambda$  of full column rank  $q$ , and write  $\Lambda = X^\top T$ . Let  $G$  be a generalized inverse of  $X^\top X$ ,  $\hat{\beta} = GX^\top y$ ,  $W = \Lambda^\top G \Lambda$  and  $u = \Lambda^\top \hat{\beta} - d$ . Then  $W$  is positive definite, and:

- (a)  $\hat{\beta}_H = \hat{\beta} - G\Lambda W^{-1}u$  satisfies  $\Lambda^\top \hat{\beta}_H = d$  and minimizes  $\|y - Xb\|^2$  subject to  $\Lambda^\top b = d$ . The set of all minimizers is  $\hat{\beta}_H + (\mathcal{N}(X) \cap \mathcal{N}(\Lambda^\top))$ .
- (b) The restricted fitted vector is unique:  $X\hat{\beta}_H = My - MTW^{-1}u = Xb_0 + P_{S_0}(y - Xb_0)$ , and  $\|y - X\hat{\beta}_H\|^2 = \text{SSE} + \text{SS}_H$ , with  $\text{SS}_H = u^\top W^{-1}u$ .
- (c) With  $\hat{\theta} = W^{-1}u$ , the restricted estimate solves the constrained normal equations (compare [Theorem 6.4.2](#))

$$X^\top X\hat{\beta}_H + \Lambda\hat{\theta} = X^\top y, \quad \Lambda^\top \hat{\beta}_H = d,$$

$$\text{and } \text{SS}_H = \hat{\theta}^\top W\hat{\theta}.$$

*Proof.*  $W = T^\top XGX^\top T = T^\top MT$  by [Theorem 6.3.7](#). It is symmetric, and positive definite by [Theorem 9.2.4\(a\)](#). Also  $XG\Lambda = XGX^\top T = MT$ .

- (a)  $\Lambda^\top \hat{\beta}_H = \Lambda^\top \hat{\beta} - WW^{-1}u = d$ . For any feasible  $b$ , [Eq. 6.3.1](#) with  $u = Xb$  gives  $\|y - Xb\|^2 = \text{SSE} + \|X\hat{\beta} - Xb\|^2$ . Write  $b = \hat{\beta}_H + v$  with  $\Lambda^\top v = 0$ . Then  $X\hat{\beta} - Xb = MTW^{-1}u - Xv$ , and the cross term vanishes:

$$(MTW^{-1}u)^\top Xv = u^\top W^{-1}T^\top Xv = u^\top W^{-1}\Lambda^\top v = 0,$$

using  $MX = X$ . Hence  $\|y - Xb\|^2 = \text{SSE} + \|MTW^{-1}u\|^2 + \|Xv\|^2$ , which is smallest exactly when  $Xv = 0$ . The minimizers are  $\hat{\beta}_H + v$  with  $v \in \mathcal{N}(X) \cap \mathcal{N}(\Lambda^\top)$ .

- (b)  $X\hat{\beta}_H = X\hat{\beta} - XG\Lambda W^{-1}u = My - MTW^{-1}u$ , and both terms are unique by [Theorem 9.2.4\(a\)](#). The minimum value is  $\text{SSE} + u^\top W^{-1}T^\top MTW^{-1}u = \text{SSE} + u^\top W^{-1}u$ , and  $u^\top W^{-1}u = \text{SS}_H$  by [Theorem 11.3.2\(a\)](#). The fitted vector  $X\hat{\beta}_H$  is the point of  $Xb_0 + S_0$  nearest to  $y$ , which is  $Xb_0 + P_{S_0}(y - Xb_0)$  by the argument in the proof of [Theorem 11.3.2\(b\)](#).
- (c)  $X^\top X\hat{\beta}_H = X^\top X\hat{\beta} - X^\top XG\Lambda W^{-1}u = X^\top y - X^\top MT\hat{\theta} = X^\top y - \Lambda\hat{\theta}$ , because  $X^\top X\hat{\beta} = X^\top y$  and  $X^\top M = X^\top$ . The last identity is  $u^\top W^{-1}u = \hat{\theta}^\top W\hat{\theta}$ .

■

Part (c) identifies  $\hat{\theta}$  as a vector of Lagrange multipliers (compare [Proposition 1.11.3](#)). Each multiplier measures how hard the data push against its constraint, and the hypothesis sum of squares is their squared Mahalanobis length. This is the “Lagrange multiplier” form of the test met again in [Section 11.4](#).

### 11.3.4 Nonzero right-hand sides: offsets

When  $\mathbf{d} \neq \mathbf{0}$  the hypothesis restricts the mean to the *affine* set  $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ , not to a subspace. [Theorem 11.3.2\(c\)](#) shows how to handle it: subtract a known point of the set from the data. The reduced model becomes

$$E(\mathbf{Y} - \mathbf{X}\mathbf{b}_0) \in \mathcal{S}_0,$$

a linear model for the adjusted response  $\mathbf{Y} - \mathbf{X}\mathbf{b}_0$  with a **known offset**  $\mathbf{X}\mathbf{b}_0$ . Any solution  $\mathbf{b}_0$  of  $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$  may be used. Two solutions differ by an element of  $\mathcal{N}(\mathbf{\Lambda}^\top)$ , so their offsets differ by an element of  $\mathcal{S}_0$ , which the reduced model absorbs. In practice one fits  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$  on a basis of  $\mathcal{S}_0$ , for example the columns of  $\mathbf{X}\mathbf{U}$  where the columns of  $\mathbf{U}$  span  $\mathcal{N}(\mathbf{\Lambda}^\top)$ , and takes the difference of residual sums of squares. The formula [Eq. 11.3.1](#) gives the same number with no second fit.

#### Example. (Regional hypotheses with an over-parameterized model)

Return to the state murder data, now with an intercept *and* an indicator for each of the four regions, so that  $\boldsymbol{\beta} = (\mu, \alpha_N, \alpha_M, \alpha_S, \alpha_W, \beta_{\text{pov}}, \beta_{\text{sin}}, \beta_{\text{urb}})^\top$ . The model matrix is  $50 \times 8$  of rank 7. No region effect is estimable on its own, but every contrast among them is.

*All four regions equal.* Write the hypothesis with all six pairwise differences  $\alpha_i - \alpha_j$  as the columns of  $\mathbf{\Lambda}$ . They are consistent and have rank  $q = 3$ . With the Moore–Penrose inverses of  $\mathbf{X}^\top \mathbf{X}$  and of  $\mathbf{W}$ , [Eq. 11.3.1](#) gives  $\text{SS}_H = 14.83$  and  $F = 2.440$ , exactly the reduced-model test of Example (Do regions matter?), as [Theorem 11.3.2\(b\)](#) requires. A second generalized inverse, made by deleting the row and column of  $\alpha_W$  from  $\mathbf{X}^\top \mathbf{X}$  before inverting, gives a different  $\hat{\boldsymbol{\beta}}$  but the same  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ ,  $\mathbf{W}$  and  $F$ .

*Midwest, South and West equal.* With  $\mathbf{\Lambda}$  made of  $\alpha_M - \alpha_S$  and  $\alpha_S - \alpha_W$ , the sum of squares is 3.881 on 2 degrees of freedom,  $F = 0.958$ ,  $p = 0.392$ . The restricted fit of [Proposition 11.3.3](#) coincides with the fit of the model in which the three regions are merged. Under the constraint, the Northeast lies 1.379 below the common level of the other three, and the coefficient of single parenthood moves from 0.428 to 0.441.

*A nonzero right-hand side.* The contrast  $\alpha_N - (\alpha_M + \alpha_S + \alpha_W)/3$ , the Northeast against the average of the other regions, is estimated as  $\hat{\psi} = -1.426$  with standard error 0.595. To test whether it equals  $-1$ , take the offset  $\mathbf{b}_0 = (0, -1, 0, \dots, 0)^\top$  and fit  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$  on a basis of  $\mathcal{S}_0$ . The drop in residual sum of squares is 1.040, equal to  $(\hat{\psi} + 1)^2 / \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$  as [Eq. 11.3.1](#) predicts, and  $F = 0.513$  with  $p = 0.478$ . Any other solution of  $\boldsymbol{\lambda}^\top \mathbf{b}_0 = -1$  gives the same answer.

*A constraint that is not a hypothesis.* The constraint  $\alpha_N = 0$  is not estimable. Minimizing the residual sum of squares subject to it gives exactly SSE: the constrained model spans the whole of  $\mathcal{C}(\mathbf{X})$ , in agreement with [Proposition 11.3.1](#).

```
import numpy as np
import statsmodels.api as sm
from scipy import linalg, stats

data = sm.datasets.statelcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWSWWSNSSSWMMMMSSNSNMMSMWNNWNSMMSWNSMSSWNSWSMW"
region = np.array(list(codes))
```

```

y = data["murder"].to_numpy()
n = len(y)
D = np.column_stack([(region == g).astype(float) for g in "NMSW"]) # all four
    regions
X = np.column_stack([np.ones(n), D, data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X) # 7, not 8

G = np.linalg.pinv(X.T @ X) # one generalized inverse of X'X
b = G @ X.T @ y # one least squares solution
s2 = np.sum((y - X @ b) ** 2) / (n - r)

def glh_test(Lam, d):
    """F test of the testable hypothesis Lam' beta = d, via any g-inverse of Lam' G
    Lam."""
    u = Lam.T @ b - d
    W = Lam.T @ G @ Lam
    q = np.linalg.matrix_rank(Lam)
    ss_h = u @ np.linalg.pinv(W) @ u
    F = ss_h / q / s2
    return ss_h, q, F, stats.f.sf(F, q, n - r)

def contrast(pairs):
    """Columns e_i - e_j on the region coefficients (positions 1..4 = N, M, S, W).
    """
    L = np.zeros((8, len(pairs)))
    for k, (i, j) in enumerate(pairs):
        L[1 + i, k], L[1 + j, k] = 1.0, -1.0
    return L

all_pairs = contrast([(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)]) # rank 3,
    redundant
three_equal = contrast([(1, 2), (2, 3)]) # M = S = W
for name, L in (("four regions equal", all_pairs), ("M = S = W", three_equal)):
    ss_h, q, F, p = glh_test(L, np.zeros(L.shape[1]))
    print(f"{name:20s} SS_H = {ss_h:.3f} q = {q} F = {F:.3f} p = {p:.4f}")

```

[Run this code online](#)

The offset computation fits the reduced model directly, as a check on the formula.

```

def sse(Z, target):
    coef, *_ = np.linalg.lstsq(Z, target, rcond=None)
    return np.sum((target - Z @ coef) ** 2)

sse_full = sse(X, y)
lam = np.array([0, 1, -1 / 3, -1 / 3, -1 / 3, 0, 0, 0]) # Northeast minus the
    other three
d = -1.0 # hypothesized difference
b0 = np.array([0, -1.0, 0, 0, 0, 0, 0, 0]) # one solution of lam' b0
    = d
U = linalg.null_space(lam[None, :]) # beta - b0 ranges over C(
    U)
sse_H = sse(X @ U, y - X @ b0) # reduced model, offset X
    b0
est = lam @ b
se = np.sqrt(s2 * lam @ G @ lam)
print(f"estimate {est:.3f} (se {se:.3f}); SSE_H - SSE = {sse_H - sse_full:.3f}")
print(f"formula: {(est - d) ** 2 / (lam @ G @ lam):.3f}")

```

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The restricted estimates, from two different generalized inverses:

```
keep = [0, 1, 2, 3, 5, 6, 7]
G2 = np.zeros((8, 8))
G2[np.ix_(keep, keep)] = np.linalg.inv((X.T @ X)[np.ix_(keep, keep)])

def restricted(Lam, d, Ginv):
    """Least squares under the testable constraint Lam' b = d (Lam of full column
    rank)."""
    bh = Ginv @ X.T @ y
    W = Lam.T @ Ginv @ Lam
    return bh - Ginv @ Lam @ np.linalg.solve(W, Lam.T @ bh - d)

bH = restricted(three_equal, np.zeros(2), G)
bH2 = restricted(three_equal, np.zeros(2), G2)
print("constraint holds:", np.allclose(three_equal.T @ bH, 0))
print("fitted values agree:", np.allclose(X @ bH, X @ bH2))
```

[Run this code online](#)

*Warning* (Hypotheses chosen after seeing the data). The same Northeast contrast, tested against zero, gives  $t = -2.398$  and  $p = 0.021$ . That p-value would be valid had the contrast been specified before the data were examined. It was not: it was suggested by the estimates, after the joint test of all regional differences had failed to reject. Among the many contrasts one could have tested, the one that looks largest is, by construction, the one most likely to look significant by chance. [Chapter 13](#) gives procedures, notably Scheffé's, that remain valid when the contrast is chosen by looking at the data.

## 11.3.5 Exercises

### 11.3.5.1 A. Check your understanding

#### Exercise A1.

In the one-way layout  $E(Y_{ij}) = \mu + \alpha_i$ ,  $i = 1, 2, 3$ , with all groups nonempty, decide which of the following are testable hypotheses: (i)  $\alpha_1 = \alpha_2$ ; (ii)  $\alpha_1 = 0$ ; (iii)  $\alpha_1 + \alpha_2 + \alpha_3 = 0$ ; (iv)  $\mu + \alpha_1 = 5$ ; (v)  $\alpha_1 - \alpha_2 = 1$ ,  $\alpha_2 - \alpha_3 = 2$ ,  $\alpha_1 - \alpha_3 = 3$ ; (vi)  $\alpha_1 - \alpha_2 = 1$ ,  $\alpha_2 - \alpha_3 = 2$ ,  $\alpha_1 - \alpha_3 = 4$ . Give  $q$  for those that are.

*Solution.* The estimable functions are the combinations  $c_0\mu + \sum_i c_i\alpha_i$  with  $c_0 = \sum_i c_i$  (Example (Four answers, one fit) and [Section 8.2](#)). (i) is testable,  $q = 1$ . (ii) and (iii) are not estimable: their coefficient of  $\mu$  is 0 while the  $\alpha$  coefficients sum to 1 and 3. They are side conditions (Theorem 8.4.2). (iv) is testable,  $q = 1$ . (v) is testable with  $q = 2$ : the three differences are estimable, have rank 2, and the right-hand sides are consistent because  $1 + 2 = 3$ . (vi) is estimable but inconsistent, since  $(\alpha_1 - \alpha_2) + (\alpha_2 - \alpha_3) = \alpha_1 - \alpha_3$  would need  $1 + 2 = 4$ . No parameter satisfies it.

### 11.3.5.2 B. Practice

**Exercise B1.**

Let  $\Lambda$  be  $p \times m$  and  $\mathbf{A}$  a nonsingular  $m \times m$  matrix. Show that the hypotheses  $\Lambda^\top \beta = \mathbf{d}$  and  $(\Lambda\mathbf{A})^\top \beta = \mathbf{A}^\top \mathbf{d}$  are the same, and that Eq. 11.3.1 gives the same  $SS_H$  for both. Conclude that the test does not depend on how a set of constraints is written.

*Solution.*  $(\Lambda\mathbf{A})^\top \mathbf{b} = \mathbf{A}^\top \mathbf{d} \iff \mathbf{A}^\top (\Lambda^\top \mathbf{b} - \mathbf{d}) = \mathbf{0} \iff \Lambda^\top \mathbf{b} = \mathbf{d}$ , since  $\mathbf{A}^\top$  is nonsingular. So the constraint sets, and by Theorem 11.3.2(b) the sums of squares, agree. Directly: with  $\mathbf{u} = \Lambda^\top \hat{\beta} - \mathbf{d}$ , the new departure is  $\mathbf{A}^\top \mathbf{u}$  and the new middle matrix is  $\mathbf{A}^\top \mathbf{W} \mathbf{A}$ , and one generalized inverse of it is  $\mathbf{A}^{-1} \mathbf{W}^{-1} \mathbf{A}^{-\top}$ , which gives  $\mathbf{u}^\top \mathbf{A} \mathbf{A}^{-1} \mathbf{W}^{-1} \mathbf{A}^{-\top} \mathbf{A}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$ . The choice of generalized inverse does not matter by Theorem 11.3.2(a).

**Exercise B2.**

In Proposition 11.3.3, show that  $E(\mathbf{X}\hat{\beta}_H) = \mathbf{X}\beta - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\Lambda^\top \beta - \mathbf{d})$  and  $\text{Cov}(\mathbf{X}\hat{\beta}_H) = \sigma^2 \mathbf{P}_{S_0}$ , where  $\mathbf{P}_{S_0} = \mathbf{M} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}\mathbf{T}^\top \mathbf{M}$ . Conclude that restricted fitted values are unbiased when  $H$  holds, have smaller variance than  $\mathbf{M}\mathbf{Y}$  in every direction, and are biased when  $H$  fails.

*Solution.*  $\mathbf{X}\hat{\beta}_H = \mathbf{M}\mathbf{Y} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{T}^\top \mathbf{M}\mathbf{Y} - \mathbf{d})$ , using  $\Lambda^\top \hat{\beta} = \mathbf{T}^\top \mathbf{M}\mathbf{Y}$ . Its mean is  $\mathbf{X}\beta - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{T}^\top \mathbf{X}\beta - \mathbf{d})$ , and  $\mathbf{T}^\top \mathbf{X}\beta = \Lambda^\top \beta$ . The random part is  $(\mathbf{M} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}\mathbf{T}^\top \mathbf{M})\mathbf{Y}$ . The matrix in parentheses is the projection onto  $\mathcal{C}(\mathbf{X})$  minus the projection onto  $\mathcal{C}(\mathbf{M}\mathbf{T})$  (Theorem 6.3.7, since  $\mathbf{W} = \mathbf{T}^\top \mathbf{M}\mathbf{T}$ ), which is  $\mathbf{P}_{S_0}$  by Theorem 6.5.2. It is symmetric idempotent, so the covariance is  $\sigma^2 \mathbf{P}_{S_0}$ . The difference  $\sigma^2(\mathbf{M} - \mathbf{P}_{S_0})$  from  $\text{Cov}(\mathbf{M}\mathbf{Y})$  is a projection times  $\sigma^2$ , so it is nonnegative definite. The bias is  $-\mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\Lambda^\top \beta - \mathbf{d})$ , zero iff  $H$  holds, because  $\mathbf{M}\mathbf{T}$  has full column rank.

**Exercise B3.**

In simple regression  $E(Y_i) = \beta_0 + \beta_1 x_i$ , derive the  $F$  statistic for  $\beta_1 = 1$  in two ways: from Eq. 11.3.1, and by regressing  $y_i - x_i$  on an intercept and comparing with the full fit. Show that both give  $F = (\hat{\beta}_1 - 1)^2 S_{xx} / s^2$ .

**Exercise B4.**

Show that the hypothesis  $\mathbf{X}\beta = \mathbf{X}\beta_0$ , with  $\beta_0$  known, is testable, that  $q = r$ , and that its  $F$  statistic is  $\{\|\mathbf{M}(\mathbf{y} - \mathbf{X}\beta_0)\|^2 / r\} / s^2$ . What is the reduced model?

**11.3.5.3 C. Going deeper****Exercise C1.**

In the one-way layout with two groups of sizes 2 and 3, take  $\Lambda = (0, 1, 0)^\top$ , so that the “hypothesis” is  $\alpha_1 = 0$ . Evaluate  $(\Lambda^\top \hat{\beta})^2 / (\Lambda^\top \mathbf{G} \Lambda)$  for two different generalized inverses  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$  and show that the values differ. Explain which step of the proof of Theorem 11.3.2 fails.

**Exercise C2.**

Let  $\mathbf{X}$  have full column rank and write  $\mathbf{\Lambda}^\top = [\mathbf{\Lambda}_1^\top, \mathbf{\Lambda}_2^\top]$  with  $\mathbf{\Lambda}_2^\top$  a nonsingular  $q \times q$  block, after permuting the coefficients so that  $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top)^\top$  conformably. Show that  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  is equivalent to  $\boldsymbol{\beta}_2 = -\mathbf{\Lambda}_2^{-\top} \mathbf{\Lambda}_1^\top \boldsymbol{\beta}_1$ , that the reduced model matrix is  $\mathbf{X}_1 - \mathbf{X}_2 \mathbf{\Lambda}_2^{-\top} \mathbf{\Lambda}_1^\top$ , and that this matrix has full column rank  $p - q$ . This is how the reduced model is fitted when no offset or null-space basis is at hand.